

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2021 - 2022)

JEE(Advanced)

UNIT TEST

14-03-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,D	A,B,C	A,D	A,B,C,D	A,B	A,D	B	B	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	50.00	0.25	6.00	5.00	0.40	0.50	0.50	0.25		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,D	A,B,C,D	A,C,D	B,C,D	A,B,C,D	A,B,C,D	D	B	D	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	6.00	0.66 or 0.67	2.00	4.00	5.00	6.00	374.15	2.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	A,C,D	B	A,B,C	A	C	C	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	2.23 or 2.24	16.33	1.33	0.00	3.14	1.41	0.14	1.57		

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JEE(Main)

UNIT TEST

14-03-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	C	B	C	D	D	C	A	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	D	D	A	B	C	D	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3.00	2.00	75.00	7.00	1.73	2.00	3.00	5.40	6.93	13.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	B	C	D	C	D	B	C	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	D	B	A	D	A	B	D	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	9.20	13.00	2.00	5.00	54.00	5.00	4.00	5.00	4.00	3.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	B	C	B	C	C	B	B	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	B	C	D	B	C	B	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.60	2.00	0.75	1.00	3.00	1.00	0.25	7.00	5.00	5.00

PAPER-1**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A, B, D)**

$$\text{Sol. } T = 2\pi\sqrt{\frac{m}{k}} \Rightarrow f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

all are independent of amplitude,

but Max. K.E. = $\frac{1}{2}m\omega^2 a^2 \rightarrow$ depends on amplitude2. **Ans. (A,B,C)**

$$\text{Sol. } \delta_{\min} = 2i - A$$

$$60 = 2(60) - A$$

$$\Rightarrow A = 60^\circ$$

$$\delta = i + e - A$$

$$65 = i + 70 - 60$$

$$\Rightarrow i = 55^\circ$$

$$\mu = \frac{\sin\left(\frac{A + \delta_{\min}}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \mu = \sqrt{3}$$

Graph is asymmetric. Hence it is not a parabola.

3. **Ans. (A,D)**

$$\text{Sol. } u = -\frac{R}{4}, f = -\frac{R}{2}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{4}{R} = \frac{-2}{R}$$

$$\Rightarrow v = +\frac{R}{2} \text{ (Right of mirror)}$$

$$m = -\frac{v}{u} = -\frac{(R/2)}{(-R/4)} = +2$$

 $\Rightarrow$ upright & magnified4. **Ans. (A,B,C,D)**

$$\text{Sol. } u_x = 40 \text{ m/s}, u_y = 40 \text{ m/s}$$

$$x = 40t = 80 \text{ m}$$

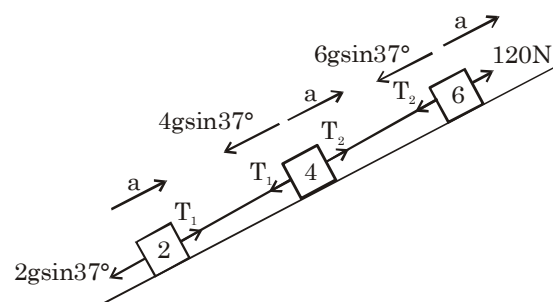
$$y = 40t - \frac{1}{2}10t^2 = 60 \text{ m}$$

$$\Rightarrow v_y = 40 - gt = 20 \text{ m/s}$$

$$\tan\theta = \frac{v_x}{v_y} = 2 \Rightarrow \theta = \tan^{-1}(2) \text{ with vertical.}$$

5. **Ans. (A, B)**

Sol.



$$120 - 6g\sin 37^\circ - T_2 = 6a$$

$$T_2 - T_1 - 4g\sin 37^\circ = 4a$$

$$T_1 - 2g\sin 37^\circ = 2a$$

$$120 - 72 = 12a$$

$$a = 4 \text{ m/s}^2$$

$$\Rightarrow T_1 = 20 \text{ N}, T_2 = 60 \text{ N}$$

6. **Ans. (A, D)**

Sol. For plane mirror, speed of image = speed of object given mirror is at rest.

7. **Ans. (B)**

Sol. (A) Eliminating 't' from both equations

$$\Rightarrow t = \frac{x}{\alpha} \Rightarrow y = x\left(1 - \frac{\beta}{\alpha}x\right)$$

$$(C) \frac{dy}{dx} = 0 = 1 - \left(\frac{2\beta}{\alpha}\right)x$$

$$\therefore x = \frac{\alpha}{2\beta}$$

8. Ans. (B)

Sol. (a) $\vec{r} = \alpha t \hat{i} + \alpha t(1 - \beta t) \hat{j}$

$$\vec{V} = \alpha \hat{i} + \alpha(1 - 2\beta t) \hat{j}$$

$$|\vec{V}| = \alpha \sqrt{1 + (1 - 2\beta t)^2}$$

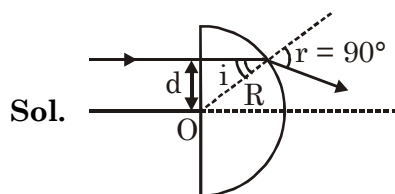
$$\text{Speed at } t = \frac{1}{4\beta}$$

$$|\vec{V}| = \alpha \sqrt{1 + \frac{1}{4}} = \frac{\sqrt{5}}{2} \alpha$$

(c) $\vec{v} = \alpha \hat{i} + \alpha(1 - 2\beta t) \hat{j}$

$$\vec{a} = -2\alpha\beta \hat{j}$$

9. Ans. (A)



As d increases, ' i ' increases,
For d to be maximum, ' i ' should be
maximum i.e., equal to critical angle
 $\Rightarrow r = 90^\circ$

$$\Rightarrow \mu \sin i = 1 \sin 90^\circ$$

$$\sqrt{2} \sin i = 1$$

$$\sin i = \frac{1}{\sqrt{2}}$$

$$\Rightarrow i = 45^\circ$$

$$\sin i = \frac{d}{R}$$

$$\Rightarrow d_{\max} = \frac{R}{\sqrt{2}}$$

10. Ans. (A)

Sol. For TIR

$$d > \frac{R}{\sqrt{2}}$$

$$\Rightarrow d_{\min} = \frac{R}{\sqrt{2}} \text{ (just)}$$

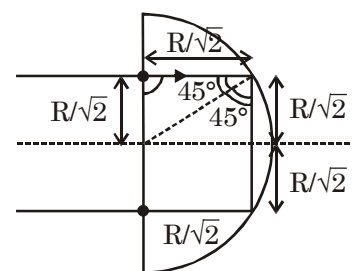
$$\mu = \frac{C}{V} = \sqrt{2}$$

$$\Rightarrow V = \frac{C}{\sqrt{2}}$$

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$= \frac{4 \cdot \left(\frac{R}{\sqrt{2}} \right)}{\left(\frac{C}{\sqrt{2}} \right)}$$

$$t = \frac{4R}{C}$$



SECTION-II

1. Ans. 50.00

Sol. $F_{\max} = m\omega^2 A = 50\sqrt{2}$

$$KE = PE \text{ at } X = \frac{A}{\sqrt{2}}$$

$$F = \frac{m\omega^2 A}{\sqrt{2}} = 50$$

2. Ans. 0.25

Sol. $\delta_a = \left(\frac{3}{2} - 1 \right) \times A = \frac{A}{2}$

$$\delta_w = \left(\frac{3/2}{4/3} - 1 \right) A = \frac{A}{8}$$

$$\frac{\delta_{\text{water}}}{\delta_{\text{air}}} = \frac{1}{4}$$

3. Ans. 6.00

Sol. For maximum speed,

$$\frac{dv}{dt} = 0 \Rightarrow a = 0$$

If at maximum speed, elongation is x , then

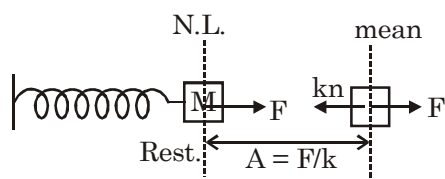
$$F = kx \Rightarrow x = \frac{F}{k}$$

Now, by WET,

$$-\frac{1}{2} kx^2 + Fx = \frac{1}{2} mv^2$$

$$\therefore V = \frac{F}{\sqrt{mk}} = 6 \text{ m/s}$$

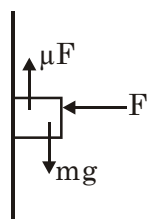
Alternate Solution



$$v_{\max} = A\omega = \left(\frac{F}{k}\right)\sqrt{\frac{k}{m}} = \frac{F}{\sqrt{km}}$$

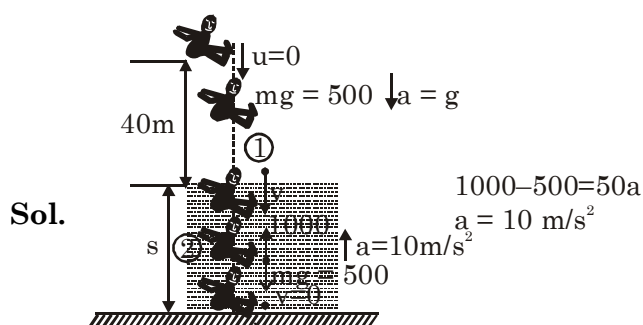
4. Ans. 5.00

Sol. $\mu F = mg$
(F = minimum force)



$$F = \frac{mg}{\mu} = \frac{0.2\text{kg} \times 10\text{m/s}^2}{0.4} = 5\text{N}$$

5. Ans. 0.40



Sol.

For (1)

$$v^2 = u^2 + 2as$$

$$v^2 = 0 + 2(10)(40)$$

$$\Rightarrow v^2 = 800$$

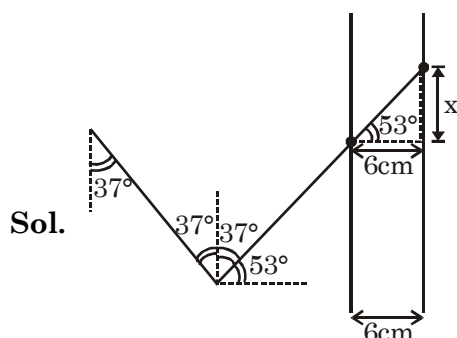
For (2)

$$v^2 = u^2 + 2as$$

$$0 = 800 + 2(-10)s$$

$$\Rightarrow s = 40\text{ m}$$

6. Ans. 0.50



Sol.

$$\tan 53^\circ = \frac{4}{3} = \frac{x}{6}$$

$$\Rightarrow x = 8\text{ cm}$$

7. Ans. 0.50

$$\text{Sol. } P_{\text{air}} = 2 = \left(\frac{3}{2} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \dots\dots(1)$$

$$P = \left(\frac{3/2}{4/3} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \dots\dots(2)$$

$$(1) \div (2)$$

$$\Rightarrow \frac{2}{P} = 4$$

$$\Rightarrow P = 0.5\text{ D}$$

8. Ans. 0.25

Sol. In first case,

$$T = 2\pi\sqrt{\frac{m}{K}} \text{ and } A = \frac{mg}{K}$$

$$\therefore A.T. = 2\pi\sqrt{\frac{m}{K}} \cdot \frac{mg}{K} = 8 \dots\dots(i)$$

In second case,

$$T_1 = 2\pi\sqrt{\frac{m}{K_1 + K_2}}$$

$$A_1 = \frac{mg}{K_1 + K_2}$$

$$\therefore A_1 T_1 = 2\pi\sqrt{\frac{m}{K_1 + K_2}} \cdot \frac{mg}{K_1 + K_2}$$

$$= 2\pi\sqrt{\frac{m}{(1+n)^2} \cdot \frac{n}{K}} \cdot \frac{mg \cdot n}{(1+n)^2 K}$$

$$= 2\pi\sqrt{\frac{m}{K}} \cdot \frac{mg}{K} \cdot \frac{n^{3/2}}{(1+n)^3}$$

$$\text{If } n = 1 \Rightarrow A_1 T_1 = 8 \cdot \frac{1}{2^3} = 1$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. **Ans.(A,B,D)**

Sol. (A) $\text{Bi}^{3+} > \text{Bi}^{5+}$ (Inert pair effect)

(B) $\text{SO}_3^{+6} > \text{SO}_2^{+4}$ (Acidic nature)

(C) $\text{Tl}^{3+} > \text{In}^{3+}$ (+1 of Tl is more stable)

(D) $\text{MgO} > \text{CaO}$ (Inter ionic distance is less in MgO)

2. **Ans. (A,B,C,D)**

3. **Ans.(A,C,D)**

When $x = 1$, liquid is pure 'A' and

$$P = P_A^0 = 100 \text{ cm Hg}$$

When $x = 0$, liquid is pure 'B' and

$$P = P_B^0 = 60 \text{ cm Hg}$$

$$\text{Now, } y = \frac{P_A}{P_{\text{total}}} = \frac{x \times P_A^0}{P_{\text{total}}} = \frac{0.5 \times 100}{(60 + 40 \times 0.5)}$$

$$= 0.625$$

$$\text{and, } \frac{1}{P_{\text{total}}} = \frac{y}{P_A^0} + \frac{1-y}{P_B^0} = \frac{0.5}{100} + \frac{1-0.5}{60}$$

$$\therefore P_{\text{total}} = P = 75 \text{ cm Hg}$$

4. **Ans.(B,C,D)**

5. **Ans.(A,B,C,D)**

6. **Ans.(A,B,C,D)**

7. **Ans.(D)**

Volume of container

$$= 1900 \times 200 \times 100 \times 10^{-3} \text{ cm}^3 = 38 \text{ litre}$$

using : $PV = nRT$ for dry NO

$$\left(\frac{760 - 40}{760} \right) \times 38 = n_{\text{NO}} \times 0.08 \times 300$$

$$\Rightarrow n_{\text{NO}} = 1.5 \text{ mole}$$

$$\text{moles of CuS} = \frac{3}{2} \times 1.5 = 2.25 \text{ moles}$$

$$\text{wt. of CuS} = 2.25 \times 95.5 = 214.875 \text{ gm}$$

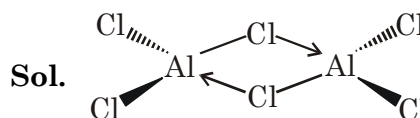
8. **Ans.(B)**

9. **Ans.(D)**

10. **Ans.(B)**

SECTION-II

1. **Ans.(6.00)**



2. **Ans.(0.66 or 0.67)**

		Position of unpaired e^-
CF_3	sp^3	sp^3 orbital
CH_3	sp^2	p orbital
NO_2	sp^2	sp^2 hybrid orbital
ClO_3	sp^3	sp^3 hybrid orbital
ClO_2	sp^2	d orbital

Sol.

$$\text{ratio} = \frac{2}{3} = 0.66 \text{ or } 0.67$$

3. **Ans. (2.00)**

4. **Ans.(4.00)**

5. **Ans. (5.00)**

6. **Ans. (6.00)**

7. **Ans.(374.15)**

$$\Delta T_b = K_b \cdot m = 0.52 \times \frac{11.7 \times 2 + \frac{34.2}{58.5}}{0.260} = 1\text{K}$$

$$\therefore \text{B.P.} = 373.15 + 1 = 374.15 \text{ K.}$$

8. **Ans.(2.00)**

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (D)**

Sol. $f(x) = \sin(3x - [3x]) = \sin(\{3x\})$

$\therefore$ period of $f(x)$ is $\frac{1}{3}$

2. **Ans. (D)**

Sol. $f(x) = \sqrt{\frac{\pi}{2} - \sin^{-1} \sqrt{\log_4 x}} - \frac{\pi}{2} + \sin^{-1} \left(\frac{1+x^2}{4x} \right)$

$$= \sqrt{-\sin^{-1} \sqrt{\log_4 x}} + \sin^{-1} \left(\frac{1+x^2}{4x} \right)$$

for defining $f(x)$:

$$-\sin^{-1} \sqrt{\log_4 x} \leq 0 \text{ \& } -1 \leq \frac{1+x^2}{4x} \leq 1$$

$$\Rightarrow \sin^{-1} \sqrt{\log_4 x} \leq 0$$

$$\therefore -1 \leq \sqrt{\log_4 x} \leq 0$$

$$\Rightarrow \sqrt{\log_4 x} = 0 \quad [x=1]$$

$\therefore$ Domain of $f(x)$ is $x = 1$

$$\Rightarrow \text{Range of } f(x) \text{ is } \left\{ \frac{\pi}{6} \right\}$$

3. **Ans. (A,C,D)**

Sol. Let common roots are α and β

$$\therefore x^3 + 5x^2 + px + q = 0 \quad \dots(1)$$

$$x^3 + 7x^2 + px + r = 0 \quad \dots(2)$$

Subtracting equation (1) - (2)

$$\therefore -2x^2 + q - r = 0 \text{ roots are } \alpha \text{ and } \beta$$

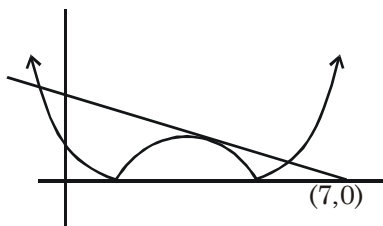
$$\therefore \alpha + \beta = 0$$

$$\text{from (1) : } \alpha + \beta + \gamma_1 = -5 \Rightarrow \gamma_1 = -5$$

$$\text{from (2) : } \alpha + \beta + \gamma_2 = -7 \Rightarrow \gamma_2 = -7$$

4. **Ans. (B)**

Sol. $|x^2 - 5x + 6| = \lambda x - 7\lambda$ has exactly three solution



If given equation has exactly three solution

$\therefore$ line $\lambda x - 7\lambda$ touches curve $-(x^2 - 5x + 6)$

$\therefore -x^2 + 5x - 6 = \lambda x - 7\lambda$ has two equal and real roots

$$x^2 + x(\lambda - 5) + 6 - 7\lambda = 0$$

$$\therefore D = 0$$

$$(\lambda - 5)^2 - 4(6 - 7\lambda) = 0$$

$$\lambda^2 + 25 - 10\lambda - 24 + 28\lambda = 0$$

$$\lambda^2 + 18\lambda + 1 = 0$$

$$\lambda = -9 + 4\sqrt{5}$$

or $\lambda = -9 - 4\sqrt{5}$ for this value of λ , line

$\lambda x - 7\lambda$ never intersect the given curve.

5. **Ans. (A,B,C)**

Sol. $g(x) = \left[\left(1 + \cos \frac{x}{50} \right)^2 - 2 \cos \frac{x}{50} - \frac{1}{2} \cos \left(\frac{x}{25} \right) - x^{101} \right]^{\frac{1}{101}}$

$$= \left[1 + \cos^2 \frac{x}{50} - \frac{1}{2} \cos \frac{x}{25} - x^{101} \right]^{\frac{1}{101}}$$

$$= \left[1 + \frac{1 + \cos \frac{x}{25}}{2} - \frac{1}{2} \cos \frac{x}{25} - x^{101} \right]^{\frac{1}{101}}$$

$$= \left[\frac{3}{2} - x^{101} \right]^{\frac{1}{101}}$$

$$\therefore g(g(x)) = \left[\frac{3}{2} - \left(\left(\frac{3}{2} - x^{101} \right)^{101} \right)^{\frac{1}{101}} \right]^{\frac{1}{101}}$$

$$= x$$

$$\therefore g(g(g(x))) = g(x)$$

6. **Ans. (A)**

Sol. $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \frac{1}{2} x |x| + \cos x + 1$$

Range of $f(x)$ is ' $\mathbb{R}$ ' $\Rightarrow$ ' f ' is ONTO

$$f(x) = \begin{cases} \frac{x^2}{2} + \cos x + 1 & , x \geq 0 \\ -\frac{x^2}{2} + \cos x + 1 & , x < 0 \end{cases}$$

$$f'(x) = \begin{cases} x - \sin x & , x > 0 \\ -x - \sin x & , x < 0 \end{cases}$$

$\therefore f'(x) > 0 \forall x \in \mathbb{R} \Rightarrow f(x)$ is one-one

Paragraph for Question 7 and 8

7. **Ans. (C)**

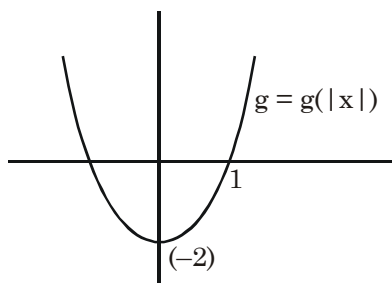
Sol. $f(x) = x^2 - 2x - 1$

$$f'(x) = 2x - 2$$

$$\therefore f'(x) \leq 0 \Rightarrow x \leq 1$$

$$\therefore a = 1 \text{ and } b = -2$$

$$g(x) = x^2 - 2x - 1 + 3x - 1 \\ = x^2 + x - 2$$



Least value of $g(x)$ is '-2'

8. **Ans. (C)**

Sol. $f : (-\infty, 1] \rightarrow [-2, \infty)$

$$y = x^2 - 2x - 1$$

$$y = (x - 1)^2 - 2$$

$$x = 1 \pm \sqrt{y + 2}$$

$$\therefore f^{-1}(x) = 1 + \sqrt{x + 2}$$

it is not possible since domain of $f(x)$ is less than 1.

$$\text{or } f^{-1}(x) = 1 - \sqrt{x + 2}$$

Paragraph for Question 9 and 10

9. **Ans. (C)**

Sol. $\therefore \frac{1}{x} + \frac{1}{x-1} + \frac{1}{x-2} = 1$ has roots a, b and c

$$\therefore x^3 - 6x^2 + 8x - 2 = 0 \text{ has roots } a, b, c$$

$$\therefore a + b + c = 6, ab + bc + ca = 8, abc = 2$$

10. **Ans. (B)**

Sol. $a^2 + b^2 + c^2 = (a + b + c)^2 - 2\Sigma ab \\ = 36 - 16 = 20$

SECTION-II

1. **Ans. 2.23 or 2.24**

Sol. $\sum_{n=1}^{\infty} \tan^{-1} \left[\frac{2n+2}{1+n(n+2)(n+1)^2} \right]$

$$\sum_{n=1}^{\infty} \tan^{-1} \left[\frac{(n+1)(n+2) - n(n+1)}{1 + (n+1)(n+2) \cdot n(n+1)} \right]$$

$$\sum_{n=1}^{\infty} [\tan^{-1} [(n+1)(n+2)] - \tan^{-1} n(n+1)]$$

$$\therefore [\tan^{-1} 6 - \tan^{-1} 2] + [\tan^{-1} 12 - \tan^{-1} 6] + \dots \\ + [\tan^{-1} (n+1)(n+2) - \tan^{-1} n(n+1)] + \dots \\ n \rightarrow \infty [\tan^{-1} (n+1)(n+2) - \tan^{-1} 2]$$

$$\therefore \frac{\pi}{2} - \tan^{-1} 2 = \cot^{-1} 2 = \sin^{-1} \frac{1}{\sqrt{5}}$$

2. **Ans. 16.33**

Sol. $\sin^{-1} \left(\frac{\sqrt{x}}{2} \right) + \sin^{-1} \left(\sqrt{1 - \frac{x}{4}} \right) + \tan^{-1} y = \frac{2\pi}{3}$

$$\sin^{-1} \frac{\sqrt{x}}{2} + \cos^{-1} \frac{\sqrt{x}}{2} + \tan^{-1} y = \frac{2\pi}{3}$$

$$\frac{\pi}{2} + \tan^{-1} y = \frac{2\pi}{3}$$

$$\tan^{-1} y = \frac{\pi}{6} \Rightarrow y = \frac{1}{\sqrt{3}}$$

$\therefore$ max. value of $(x^2 + y^2)$ occurs

$$\text{at } x = 4 \text{ and } y = \frac{1}{\sqrt{3}}$$

3. **Ans. 1.33**

Sol. Let $x = \frac{1}{t}$

$$\therefore \lim_{t \rightarrow 0} \frac{(t+1)^{2/3} - (1-t)^{2/3}}{t}$$

$$= \frac{(t+1)^{2/3} - 1}{t+1-1} + \frac{(1-t)^{2/3} - 1}{(1-t)-1}$$

$$\frac{2}{3} + \frac{2}{3} = \frac{4}{3}$$

4. **Ans. 0.00**

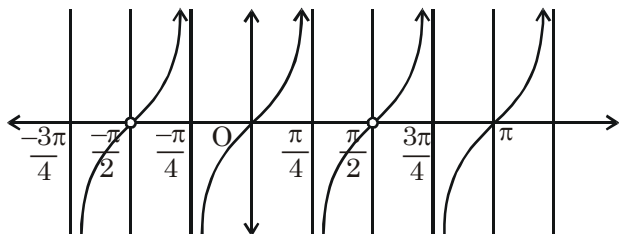
Sol. Let $x = \frac{\pi}{2} + h$

$$\therefore \lim_{h \rightarrow 0} \frac{-(1 - \cosh) \cosh}{h^2} \cdot h^2 = 0$$

5. Ans. 3.14

Sol. $f(x) = \frac{2 \sin 2x \cdot \cos x}{2 \cos 2x \cdot \cos x}$

$$f(x) = \tan 2x, \quad x \neq (2n+1)\frac{\pi}{2}$$



period is π .

6. Ans. 1.41

Sol. $a_n = \sqrt{2 + a_{n-1}}$

$$= \sqrt{2 + \sqrt{2 + a_{n-2}}} = \sqrt{2 + \sqrt{2 + \sqrt{2 + a_{n-3}}}}$$

$$\lim_{n \rightarrow \infty} a_n = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots \infty}}} = 2$$

7. Ans. 0.14

Sol. $f(0) = 0, f(1) = 2, f(2) = 1, f(3) = 2, f(4) = 3$

$$f(-5) = f(-5 + 8) = f(3) = 2$$

$$f(20) = f(2 \times 8 + 4) = f(4) = 3$$

$$f(-10) = f(10) = f(8 + 2) = f(2) = 1$$

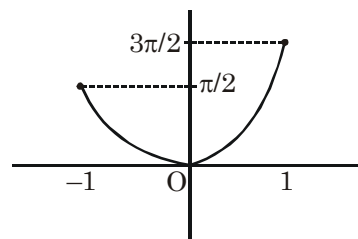
$$f(17) = f(2 \times 8 + 1) = f(1) = -2$$

$$\therefore |\tan^{-1}(\tan(2 + 3 + \cos^{-1}(1) + (-2)))|$$

$$|\tan^{-1}(\tan 3)| = 0.14$$

8. Ans. 1.57

Sol. $f(x) = \begin{cases} 3 \sin^{-1} x & , \quad 0 \leq x \leq 1 \\ -\sin^{-1} x & , \quad -1 \leq x < 0 \end{cases}$



$$k = \frac{\pi}{2}$$

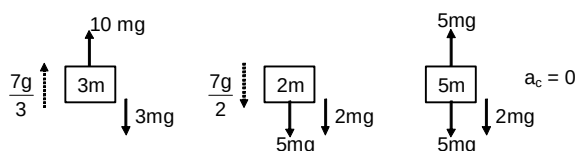
PAPER-2**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (C)**

Sol. From ground frame acceleration of the block is zero. So spring force must be balanced with the friction.

$$kx = \mu mg$$

2. **Ans. (C)**

Sol. FBD just after string is cut.

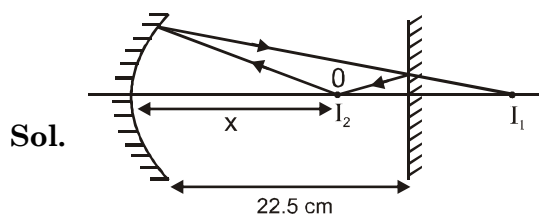
3. **Ans. (C)**

Sol. Apparent depth = $\frac{\text{Real depth}}{\mu}$

Real depth from five faces = $\mu \times 2 = 3$ cm

Real depth from sixth face = $\mu \times 4 = 6$ cm

so dimensions of cube $6 \times 6 \times 9$ cm³

4. **Ans. (B)****Sol.**

I_1 is the image formed by concave mirror.

For reflection by concave mirror

$$u = -x, \quad v = -(45 - x), \quad f = -10 \text{ cm},$$

$$\frac{1}{-10} = \frac{1}{-(45-x)} + \frac{1}{-x}$$

$$\frac{1}{10} = \frac{x+45-x}{x(45-x)} \Rightarrow x^2 - 45x + 450 = 0$$

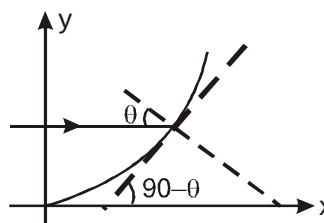
$$\Rightarrow x = 15 \text{ cm}, 30 \text{ cm}$$

but $x = 30$ cm is not acceptable because $x < 22.5$ cm.

5. **Ans. (C)**

Sol. $\sin \theta_c = \frac{1}{\mu} = \frac{3}{5}$

$$\frac{dy}{dx} = \tan(90^\circ - \theta)$$



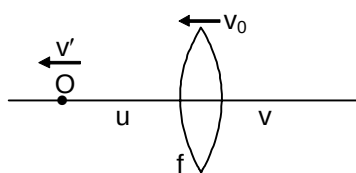
$$\cot \theta = 2x$$

$$\tan \theta = \frac{1}{2x}$$

$$\frac{3}{4} = \frac{1}{2x} \rightarrow x = \frac{2}{3} \text{ cm}, y = \frac{4}{9} \text{ cm}$$

6. **Ans. (D)**

Sol. If $\frac{\text{size of object}}{\text{size of image}} = n$



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Using sign convention and differentiating

$$-\frac{1}{v^2} \frac{dv}{dt} + \frac{1}{u^2} \frac{du}{dt} = 0$$

$$\Rightarrow \frac{du}{dt} = \frac{u^2}{v^2} \cdot \frac{dv}{dt} \Rightarrow (v' - v_0) = -n^2 \cdot v_0$$

$$\Rightarrow v' = (1 - n^2)v_0$$

7. Ans. (D)

Sol. In rarer medium ray bends away from normal.

8. Ans. (C)

Sol. $r_2 = 0$

So $r_1 = A = 30^\circ$

from snell's law

$$1 \sin i = \mu \sin r_1$$

$$\sin i = \sqrt{2} \sin 30^\circ$$

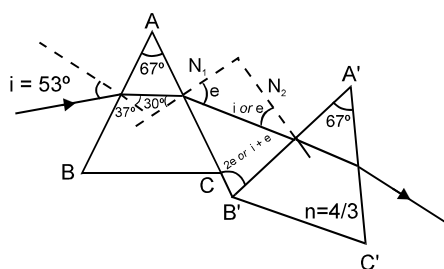
$$i = 45^\circ$$

9. Ans. (A)

Sol. Let be the angle of emergence from the first prism be 'e'

calculating step by step we get $e = \sin^{-1} \frac{2}{3}$

Then for net deviation to be double, the incident ray on side A'B' of second prism should make angles i or e with normal.

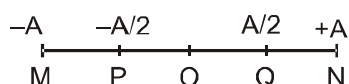


Hence the angle between the given then will be $2e$ or $i + e$.

10. Ans. (C)

Sol. It T be the time period; time to go from O

to Q is $\frac{T}{12}$ and from M to P is $\frac{T}{6}$.



The displacement is $\frac{A}{2}$ when particle goes

from O to Q, from O to N to Q,

from O to N to O to P, and so on

$$\therefore t = \frac{T}{12} \text{ or } t = \frac{T}{4} + \frac{T}{6} = \frac{5T}{12}$$

$$\text{or } t = \frac{T}{2} + \frac{T}{12} = \frac{7T}{12}$$

Hence possible time period T is

$$T = 12 \text{ s} \quad \text{or } T = \frac{12 \times 1}{5} = 2.4 \text{ s}$$

$$\text{or } T = \frac{12 \times 1}{7} \text{ s}$$

similarly, displacement is $\frac{A}{2}$ when

particle goes from M to P or M to N to P

Hence the possible time period T is

$$T = 1 \times 6 = 6 \text{ s} \quad \text{or } T = \frac{6 \times 1}{5} \text{ s} = 1.2 \text{ s}$$

Ans. $T = 1.2 \text{ s}, 6 \text{ s}, 2.4 \text{ s}, 12 \text{ s}$

11. Ans. (D)

Sol. Power supplied by the restoring force

($F = Kx$) is :

$$P = FV$$

$$P = (Kx) v_n \sqrt{A^2 - x^2}$$

$$P = K \sqrt{\frac{K}{m}} \sqrt{A^2 x^2 - x^4}$$

P will be max when $\frac{dP}{dx} = 0$

$$\Rightarrow A^2(2x) - 4x^3 = 0$$

$$x = \frac{A}{\sqrt{2}}$$

$$P_{\max} = P \Big|_{\text{at } x = \frac{A}{\sqrt{2}}} = K \sqrt{\frac{K}{m}} \sqrt{A^2 \left(\frac{A}{\sqrt{2}}\right)^2 - \left(\frac{A}{\sqrt{2}}\right)^4};$$

$$P_{\max} = \frac{K^{\frac{3}{2}} A^2}{2\sqrt{m}}$$

12. Ans. (C)

Sol. $KE = \frac{1}{2} m \omega^2 a^2 \cos^2 \omega t$ and

$$PE = \frac{1}{2} m \omega^2 a^2 \sin^2 \omega t$$

$$KE - PE = \frac{1}{2} m \omega^2 a^2 (\cos^2 \omega t - \sin^2 \omega t)$$

$$= \frac{1}{2} m \omega^2 a^2 \cos 2\omega t$$

$\therefore$ Angular frequency = 2ω

$\therefore$ The time period = $\frac{T}{2} = 2 \text{ s}$

13. Ans. (D)

Sol. Amplitude is obtained for $v = 0$

$$\therefore A = \sqrt{\frac{E}{a}}$$

Maximum velocity is obtained for $x = 0$

$$V_{\max} = \sqrt{\frac{E}{a}}$$

$$V_{\max} = A\omega$$

$$\omega = \frac{\sqrt{\frac{E}{a}}}{\sqrt{\frac{E}{a}}} = \sqrt{\frac{a}{b}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{b}{a}}$$

14. Ans. (D)

Sol. $a = -\beta(x-2)$ Compares with the standard equation of SHM $a = -\omega^2 x$.

$$\therefore \omega = \sqrt{\beta} \text{ (Equilibrium position } x = 2)$$

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\beta}}$$

15. Ans. (A)

Sol. Initially distance of suspension point from

$$CM = \ell = \frac{1}{2} \text{ m}$$

To get the same time period, we can hinge

$$\text{the rod at distance } \frac{K_{cm}^2}{\ell} = \frac{\left(\frac{1}{\sqrt{12}}\right)^2}{1/2} = \frac{1}{6} \text{ m}$$

Here ;

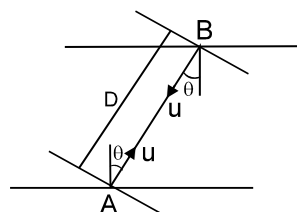
$$I_{cm} = \frac{mL^2}{12} = mK_{cm}^2 \Rightarrow K_{cm} = \frac{L}{\sqrt{12}} = \frac{1}{\sqrt{12}}$$

16. Ans. (B)

$$\text{Sol. } KA \times \left(\frac{2a}{3} - \frac{a}{2}\right) = mg \left(\frac{a}{2}\right)$$

$$A = \frac{3mg}{K}$$

17. Ans. (C)



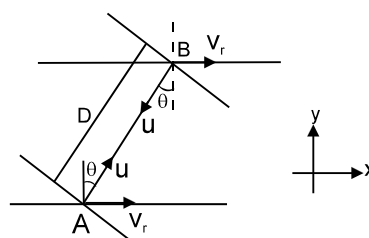
Sol. (a)

Solving with respect to river, velocity of swimmer will be 'u' towards each other. so time taken to meet

$$t = \frac{D}{2u} \text{ Ans.}$$

(b) For path to be at right angle to each other, their velocity vector with respect to ground must be at right angle

Taking axis system as shown



$$\vec{V}_A = (u \sin \theta + v_r) \hat{i} + u \cos \theta \hat{j}$$

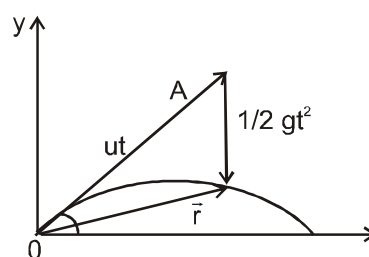
$$\vec{V}_B = (v_r - u \sin \theta) \hat{i} - u \cos \theta \hat{j}$$

For $\vec{V}_A$ & $\vec{V}_B$ to be perpendicular $\vec{V}_A \cdot \vec{V}_B = 0$

$$(u \sin \theta + v_r)(v_r - u \sin \theta) - u^2 \cos^2 \theta = 0$$

$$\Rightarrow v_r = u$$

18. Ans. (D)



Sol.

$$AB = \frac{1}{2} g(T/2)^2 = \frac{1}{8} gT^2$$

$$CD = \frac{1}{2} gT^2$$

$$CD/AB = 4$$

19. Ans. (B)

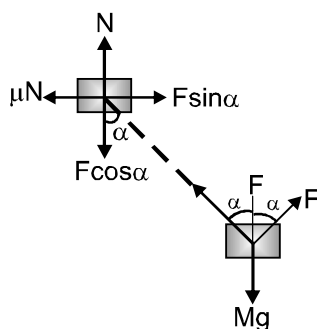
Sol. Horizontal component of velocity at a time t is $v_x = u \cos \theta$
Vertical components of velocity at that time t is $v_y = u \sin \theta - gt$
They will become equal at time $t = t^*$
if $u \cos \theta = u \sin \theta - gt^*$
which gives, $t^* = \frac{u}{g} (\sin \theta - \cos \theta)$
Now t^* must be positive. hence $\sin \theta > \cos \theta$
or $\theta > 45^\circ$. Hence the correct choice is (B).

20. Ans. (C)

Sol. Due to gravity velocity at the end of travel is high, so covers bigger displacement in later part of journey.

SECTION-II

1. Ans. 3.00



Sol.

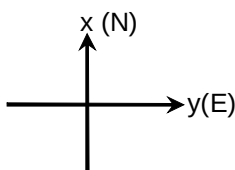
$$F \sin \alpha = \mu(F \cos \alpha)$$

$$\tan \alpha = \mu$$

$$\mu = 0.75$$

$$4\mu = 3.$$

2. Ans. 2.00



Sol.

$$\vec{a}_1 = 4\hat{j}$$

$$\vec{a}_{1/s} = -2\sqrt{3}\hat{j} + 2\hat{j}$$

$$\vec{a}_s = \vec{a}_1 - (\vec{a}_{1/s})$$

$$= 2\sqrt{3}\hat{i} + 2\hat{j}$$

$$\vec{a}_2 = 2\sqrt{3}\hat{i}$$

$$\vec{a}_{2/s} = -2\hat{j}$$

3. Ans. 75.00

Sol. $f = +30$ cm.
for second lens,
 $u = -50$ cm $v = ?$ $f = +30$
$$v = \frac{uf}{u+f} = +75 \text{ cm}$$

4. Ans. 7.00

Sol. According to question, image position for, concave mirror is 35 cm,

$$\text{size of image } (h_i) = \frac{35}{20} \times (h_o) = \frac{35}{20} (40 \text{ m})$$

$$= 7 \text{ cm}$$

5. Ans. 1.73

Sol. Given $i = 60^\circ \Rightarrow \delta = i$ ($\because e = A$)

$$\mu = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \frac{A}{2}}$$

Here angle of deviation is minimum
($\because i = e$)

$$\mu = \frac{\sin \left(\frac{60^\circ + 60^\circ}{2} \right)}{\sin (60^\circ / 2)} = \sqrt{3}$$

6. Ans. 2.00

Sol. The coordinates of the particles are
 $x_1 = A_1 \cos \omega t$, $x_2 = A_2 \cos \omega t$
separation $= x_1 - x_2 = (A_1 - A_2) \cos \omega t$
 $= 12 \cos \omega t$
Now $x_1 - x_2 = 6 = 12 \cos \omega t$

$$\Rightarrow \omega t = \frac{\pi}{3} \Rightarrow \frac{2\pi}{12} \cdot t = \frac{\pi}{3}$$

$$\Rightarrow t = 2 \text{ s Ans.}$$

7. Ans. 3.00

Sol. At $t =$

$$\text{Slope} = \frac{d \left(\frac{1}{v} \right)}{dt} = -1$$

$$\Rightarrow -\frac{1}{v^2} \frac{dv}{dt} = -1$$

$$\Rightarrow \left(\frac{1}{\sqrt{3}} \right)^2 \frac{dv}{dt} = 1 \Rightarrow a = \frac{dv}{dt} = 3 \text{ m/s}^2$$

8. Ans. 5.40

Sol. For First second

$$1.5 = u + a \times 1 \Rightarrow u + a = 1.5 \text{ - (i)}$$

For Sixth second

$$0 = u + a \times 6 \Rightarrow u = -6a \text{ - (ii)}$$

Using (i) & (ii)

$$a = -0.3 \text{ m/sec}^2$$

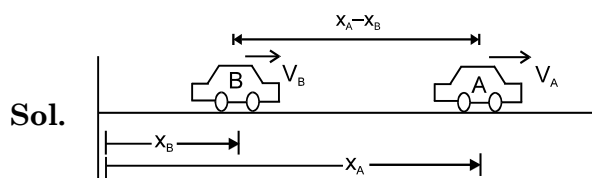
$$\& u = +1.8 \text{ m/sec}^2$$

$\therefore$ In 6 second

$$x = 1.8 \times 6 + \frac{1}{2}(-0.3) \times 6^2$$

$$= 10.8 - 5.4 = 5.4 \text{ m}$$

9. Ans. 6.93



As given

$$(V_A - V_B) \propto x_A - x_B$$

$$(V_A - V_B) = K(x_A - x_B)$$

$$\text{when } x_A - x_B = 10$$

$$\text{We have } V_A - V_B = 10$$

We get

$$10 = K10 \Rightarrow K = 1$$

$$\Rightarrow V_A - V_B = (x_A - x_B) \quad \dots(1)$$

Now Let

$$x_A - x_B = y \quad \dots(2)$$

On differentiating with respect to 't' on both side.

$$\frac{dx_A}{dt} - \frac{dx_B}{dt} = \frac{dy}{dt} \quad V_A - V_B = \frac{dy}{dt} \quad \dots(3)$$

Using (1), (2), (3)

We get

$$\frac{dy}{dt} = y$$

Here y represents separation between two cars

$$\int_{10}^{20} \frac{dy}{y} = \int_0^{t_1} dt \quad [\log_e y]_{10}^{20} = t_1$$

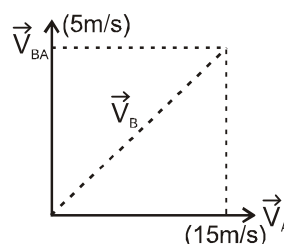
$$t_1 = (\log_e 2) \text{ sec}$$

10. Ans. 13.00

Sol. Velocity of A and velocity of B relative to A are as shown.

$$\therefore \vec{V}_B = \vec{V}_{BA} + \vec{V}_A$$

$$\Rightarrow V_0 = \sqrt{5^2 + 12^2} = 13$$



and direction of $\vec{V}_B$ is away from $\vec{V}_A$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(B)

2. Ans.(A)

3. Ans.(B)

$$\text{If the given radius ratio } \left(\frac{r_{A^+}}{r_{B^-}} \right) = (\sqrt{2} - 1),$$

this implies that cations are present in octahedral voids made by anions. B^- forms fcc lattice and A^+ are present in their octahedral voids.

Now we can say

• Anions are in contact with each other

• Cations are NOT in contact with each other.

• By face diagonal we can say.

$$\sqrt{2} a = 4r_{B^-}$$

$$\Rightarrow a = 2\sqrt{2} r_{B^-}$$

Total volume occupied by all ions

$$= 4 \left[\frac{4}{3} \pi r_{A^+}^3 + \frac{4}{3} \pi r_{B^-}^3 \right]$$

$$= \frac{6}{3} \pi (r_{A^+}^3 + r_{B^-}^3) = \frac{16\pi}{3} [5\sqrt{2} - 6]$$

$$\text{Packing fraction} = \frac{\frac{16\pi}{3} (5\sqrt{2} - 6) r_{B^-}^3}{16\sqrt{2} r_{B^-}^3}$$

$$= \frac{\pi}{3\sqrt{2}} (5\sqrt{2} - 6) = 0.79$$

4. Ans.(C)

5. Ans.(D)

6. Ans.(C)

$$k_f \propto \frac{T_f^{0.2}}{\Delta H_f} \text{ and } k_b \propto \frac{T_b^{0.2}}{\Delta H_{\text{vap.}}}$$

7. Ans.(D)

8. Ans (B)

Chiral centre present with symmetry in the molecule.

9. Ans.(C)

10. Ans.(D)

11. Ans.(B)

12. Ans(C)

13. Ans.(D)

14. Ans.(B)

15. Ans. (A)

$$PV = \frac{W}{M} RT \Rightarrow \frac{WT}{M} = \frac{PV}{R} = \text{const.}$$

$$\Rightarrow \frac{W_1 T_1}{M_1} = \frac{W_2 T_2}{M_2}$$

$$\text{or, } \frac{1.5 \times 400}{M_1} = \frac{9.6 \times 300}{48} \Rightarrow M_1 = 10$$

16. Ans.(D)

17. Ans.(A)

18. Ans.(B)

Sol: If molality and molarity of NO_3^- ion is numerically same, it is same for all components and for it,

numerical value of ml of solution = numerical value of gm of solvent

or, $1000 = 1000 d - (2 \times 164 + 3 \times 213)$
(for 1 litre solution)

$\therefore$ Density of solution, $d = 1.967 \text{ gm/ml}$

19. Ans.(D)

20. Ans. (D)

SECTION-II

1. Ans.(9.20)

2. Ans.(13.00)

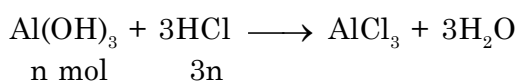
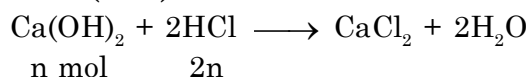
3. Ans.(2.00)

4. Ans. (5.00)

5. Ans.(54.00)

6. Ans. (5.00)

7. Ans. (4.00)



Mole of HCl = 5n

$$0.5 \times 4 = 5n$$

$$2 = 5n$$

$$n = \frac{2}{5} = 0.4 \text{ mol}$$

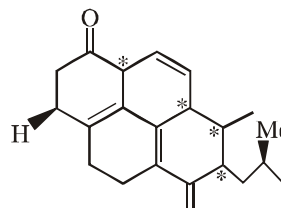
Molarity of Ca(OH)_2 in original sol

$$= \frac{\text{moles of Ca(OH)}_2}{\text{Vol of solution}}$$

$$= \frac{0.4}{100} \times 1000 = 4 \text{ M}$$

8. Ans.(5.00)

9. Ans. (4.00)



10. Ans.(3.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (B)**

Sol. $x^2 - 3x + 2 = 0$ Roots $x = 1, x = 2$
if $x = 1$ is the common root
then $1 - 4 + a = 0 \Rightarrow a = 3$
and $1 + 2 - a = 0 \Rightarrow a = 3 \quad \left. \vphantom{\begin{matrix} 1 - 4 + a = 0 \\ 1 + 2 - a = 0 \end{matrix}} \right\} \Rightarrow a = 3$
Now if $x = 2$ is the common root then
 $4 - 8 + a = 0 \Rightarrow a = 4$
and $4 + 4 - a = 0 \Rightarrow a = 8$
no value of 'a' is possible.
so, $a = 3$ is the only possible value of 'a'

2. **Ans. (D)**

Sol. $(2a - 1) f(2) < 0$
 $\Rightarrow (2a - 1)(8a - 4 + 2 - 1) < 0$
 $\Rightarrow (2a - 1)(8a - 3) < 0$

$$\Rightarrow a \in \left(\frac{3}{8}, \frac{1}{2} \right)$$

$$\text{Now } \frac{13}{12} \in \left(\frac{3}{8}, \frac{1}{2} \right) \text{ and } \frac{15}{32} \in \left(\frac{3}{8}, \frac{1}{2} \right)$$

$$\text{and } \frac{7}{16} \in \left(\frac{3}{8}, \frac{1}{2} \right) \text{ but } \frac{5}{8} \notin \left(\frac{3}{8}, \frac{1}{2} \right)$$

3. **Ans. (B)**

Sol. $x^2 - x - 1 = 0 \quad \begin{cases} \alpha \\ \beta \end{cases} \quad (\alpha, \beta \neq 0, \alpha \neq \beta)$

$$\text{Now } a\alpha^9 + b\alpha^8 + 1 = 0$$

$$\Rightarrow a\alpha + b + \frac{1}{\alpha^8} = 0 \quad \dots(1)$$

$$\text{and } a\beta^9 + b\beta^8 + 1 = 0$$

$$\Rightarrow a\beta + b + \frac{1}{\beta^8} = 0 \quad \dots(2)$$

$$a(\alpha - \beta) + \frac{1}{\alpha^8} - \frac{1}{\beta^8} = 0$$

$$\Rightarrow a(\alpha - \beta) - \frac{(\alpha - \beta)(\alpha + \beta)(\alpha^2 + \beta^2)(\alpha^4 + \beta^4)}{(\alpha\beta)^8} = 0$$

$$\Rightarrow a = \frac{(\alpha + \beta)(\alpha^2 + \beta^2)(\alpha^4 + \beta^4)}{(\alpha\beta)^8}$$

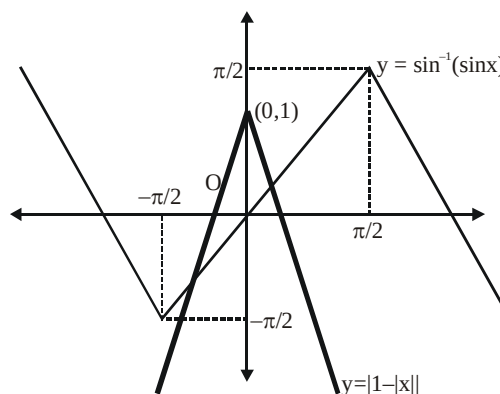
$$a = (1)(3)(7) = 21$$

4. **Ans. (C)**

$$\begin{aligned} \text{Sol. } \lim_{x \rightarrow 2} \frac{\sin((x-2)(x-3))}{(x-1)(x-2)} \\ = \lim_{x \rightarrow 2} \frac{\sin((x-2)(x-3))}{(x-2)(x-3)} \times \frac{(x-3)}{(x-1)} \\ = (1) \left(-\frac{1}{1} \right) = -1 \end{aligned}$$

5. **Ans. (B)**

Sol.



6. **Ans. (C)**

$$\text{Sol. } T_r = \tan^{-1} \left(\frac{2}{r^2} \right) = \tan^{-1} \left(\frac{(r+1) - (r-1)}{1 + (r^2 - 1)} \right)$$

$$T_r = \tan^{-1}(r+1) - \tan^{-1}(r-1)$$

$$T_1 = \tan^{-1}2 - \tan^{-1}0$$

$$T_2 = \tan^{-1}3 - \tan^{-1}1$$

$$T_3 = \tan^{-1}4 - \tan^{-1}2$$

$$T_4 = \tan^{-1}5 - \tan^{-1}3$$

$$T_{n-1} = \tan^{-1}(n) - \tan^{-1}(n-2)$$

$$T_n = \tan^{-1}(n+1) - \tan^{-1}(n-1)$$

$$\sum_{r=1}^n T_r = \tan^{-1}(n+1) + \tan^{-1}(n) - \tan^{-1}(1)$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n T_r = \frac{\pi}{2} + \frac{\pi}{2} - \frac{\pi}{4} = \frac{3\pi}{4}$$

7. Ans. (C)

$$\begin{aligned}\text{Sol. } \sin\left(\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{5}\right) \\&= \sin\left(\tan^{-1}\left(\frac{\frac{1}{3} + \frac{1}{5}}{1 - \frac{1}{3} \cdot \frac{1}{5}}\right)\right) \\&= \sin\left(\tan^{-1}\left(\frac{8}{14}\right)\right) \\&= \sin\left(\tan^{-1}\left(\frac{4}{7}\right)\right) \\&= \sin\left(\sin^{-1}\left(\frac{4}{\sqrt{65}}\right)\right) = \frac{4}{\sqrt{65}}\end{aligned}$$

8. Ans. (B)

$$\begin{aligned}\text{Sol. } 3f(x) + 2f(-x) &= x^2 - x \quad \forall x \in \mathbb{R} \\3f(-x) + 2f(x) &= x^2 + x\end{aligned}$$

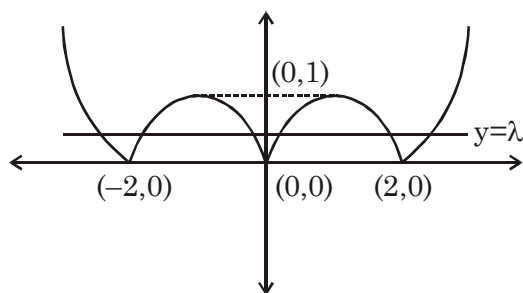
$$5f(x) = x^2 - 5x$$

$$\Rightarrow f(x) = \frac{1}{5}x^2 - x$$

$\Rightarrow f(x)$ is many one

9. Ans. (B)

Sol.



For six solutions $0 < \lambda < 1$

So possible value of $\lambda = \frac{4}{7}$

10. Ans. (B)

$$\text{Sol. } f(x) = \frac{x^3}{\sqrt{\sin^{-1}(\ln x) - \cos^{-1}(\ln x)}}$$

$$\sin^{-1}(\ln x) - \cos^{-1}(\ln x) > 0$$

$$\Rightarrow 2\sin^{-1}(\ln x) - \frac{\pi}{2} > 0$$

$$\Rightarrow \sin^{-1}(\ln x) > \frac{\pi}{4}$$

$$\Rightarrow 1 \geq \ln x > \frac{1}{\sqrt{2}}$$

$$\Rightarrow e \geq x > e^{\frac{1}{\sqrt{2}}}$$

$$\text{domain} \equiv \left[e^{\frac{1}{\sqrt{2}}}, e\right]$$

11. Ans. (C)

$$\begin{aligned}\text{Sol. } g(f(1)) + g(f(-1)) \\&= g(5) + g(1) \\&= (2(5) + 2) + (2(1) + 2) = 16\end{aligned}$$

12. Ans. (C)

$$\begin{aligned}\text{Sol. } f(x) &= x^7(x-5) - x^6(x-5) + x^3(x-5) + x(x-5) + 2 \\&\quad (x-5) + 13 \\&\Rightarrow f(5) = 13\end{aligned}$$

13. Ans. (B)

$$\text{Sol. } \lim_{x \rightarrow \infty} \left(\frac{ax^2 - ax^4 - 1 - x^4}{(1 - x^2)} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{(-a-1)x^4 + ax^2 - 1}{(1 - x^2)}$$

$$\Rightarrow -a - 1 = 0$$

$$\Rightarrow a = -1$$

$$\text{Now } L = \lim_{x \rightarrow \infty} \frac{-x^2 - 1}{1 - x^2} = 1$$

14. Ans. (C)

$$\text{Sol. } y = \tan^{-1}(-(x-1)^2 - 1)$$

$$y = -\tan^{-1}(1 + (x-1)^2)$$

$\Rightarrow f(x)$ is many one

$$\text{Now } 1 + (x-1)^2 \geq 1$$

$$\Rightarrow \frac{\pi}{2} > \tan^{-1}(1 + (x-1)^2) \geq \frac{\pi}{4}$$

$$\Rightarrow -\frac{\pi}{2} < -\tan^{-1}(1 + (x-1)^2) \leq -\frac{\pi}{4}$$

$$\Rightarrow \text{Range} \equiv \left[-\frac{\pi}{2}, -\frac{\pi}{4}\right] \Rightarrow f(x) \text{ is onto}$$

15. Ans. (D)

$$\text{Sol. } f(x) = \text{sgn}\left(1 - 2\sin^2 x - 2\sin x + \frac{1}{2}\right)$$

$$= \text{sgn}\left(-2\left(\sin^2 x + \sin x - \frac{3}{4}\right)\right)$$

$$= \text{sgn}\left(2 - 2\left(\frac{1}{2} + \sin x\right)^2\right)$$

$$\text{L.H.L} = \lim_{x \rightarrow \frac{\pi}{6}^-} \text{sgn}\left(2 - 2\left(\frac{1}{2} + \sin x\right)^2\right)$$

$$= 1$$

$$\text{R.H.L} = \lim_{x \rightarrow \frac{\pi}{6}^+} \text{sgn}\left(2 - 2\left(\frac{1}{2} + \sin x\right)^2\right)$$

$$= -1$$

$$\text{L.H.L} \neq \text{R.H.L}$$

$\Rightarrow$ 'L' does not exist

16. Ans. (B)

$$\text{Sol. } y = \frac{11x^2 + 12x + 6}{x^2 + 4x + 2}$$

$$\Rightarrow yx^2 + 4xy + 2y = 11x^2 + 12x + 6$$

$$\Rightarrow (y - 11)x^2 + 4(y - 3)x + 2y - 6 = 0$$

if $y \neq 11$

$$D \geq 0$$

$$16(y - 3)^2 - 4(y - 11)(2y - 6) \geq 0$$

$$\Rightarrow 2(y - 3)^2 - (y - 11)(y - 3) \geq 0$$

$$\Rightarrow (y - 3)(2y - 6 - y + 11) \geq 0$$

$$\Rightarrow (y - 3)(y + 5) \geq 0$$

$$\Rightarrow y \in (-\infty, -5] \cup [3, \infty) - \{11\}$$

for $y = 11$

$$11x^2 + 44x + 22 = 11x^2 + 12x + 6$$

$$\Rightarrow 32x = -16 \Rightarrow \boxed{x = -\frac{1}{2}}$$

$$\text{Range} \neq (-\infty, -5] [3, \infty)$$

$$a = -5, b = 3$$

$$a + b = -2$$

17. Ans. (C)

$$\text{Sol. } \lim_{x \rightarrow 0} \frac{\tan\left(\frac{x+x^2}{a^4}\right)}{\left(\frac{x+x^2}{a^4}\right)} \times \left(\frac{e^x - 1}{x}\right)^2 \times \frac{\frac{x^3}{2}}{\ln\left(1 + \frac{x^3}{2}\right)} \times \frac{(x+x^2)}{a^4} \times \frac{x^2}{\frac{x^3}{2}} = \frac{2}{16}$$

$$\Rightarrow (1)(1)(1) \times \frac{2}{a^4} = \frac{2}{16}$$

$$\Rightarrow a^4 = 16 \Rightarrow a = \pm 2$$

18. Ans. (B)

$$\text{Sol. } f(1) = 7 \Rightarrow f^{-1}(7) = 1$$

$$\Rightarrow g(7) = 1$$

19. Ans. (B)

$$\text{Sol. } f(x) = \{x+1\} + \{x+3\} + \{x+5\} + \dots + \{x+199\}$$

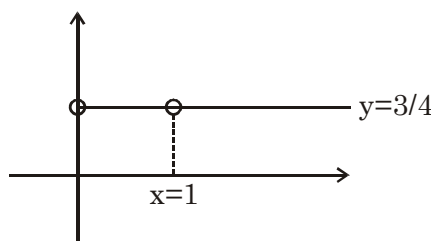
$$\Rightarrow f(x) = \{x\} + \{x\} + \{x\} + \dots + \{x\}$$

$$f(x) = 100\{x\} \Rightarrow f(\sqrt{3}) = 100(0.732) = 73.2$$

$$\Rightarrow [f(\sqrt{3})] = 73$$

20. Ans. (B)

$$\text{Sol. } f(x) = \log_{x^4} x^3 = \frac{3}{4} \forall x \in (0, \infty) - \{1\}$$



$$\text{Range} = \left\{\frac{3}{4}\right\}$$

'f' is many-one and into

SECTION-II

1. Ans. 0.60

$$\text{Sol. } a(x) = \alpha^x + \beta^x$$

$$(\alpha + \beta) a(x) = (\alpha + \beta) (\alpha^x + \beta^x)$$

$$-3a(x) = \alpha^{x+1} + \beta^{x+1} + \alpha\beta(\alpha^{x-1} + \beta^{x-1})$$

$$-3a(x) = a(x+1) + a(x-1)$$

for $x = 7$

$$-3a(7) = a(8) + a(6)$$

$$\Rightarrow \lambda = -3 \Rightarrow \frac{|\lambda|}{5} = \frac{3}{5} = 0.60 \text{ Ans.}$$

2. Ans. 2.00

$$\text{Sol. } y = \log_5(3\sin x + 4\cos x + 20)$$

maximum value of $3\sin x + 4\cos x$ is '5'

$$\Rightarrow y_{\max} = \log_5(5 + 20) = \log_5(25) = 2$$

3. Ans. 0.75

Sol. $\tan^{-1}(x^2 + 5x + 3) + \cot^{-1}(2x + a) = \frac{\pi}{2}$

$$\Rightarrow \tan^{-1}(x^2 + 5x + 3) = \frac{\pi}{2} - \cot^{-1}(2x + a)$$

$$\Rightarrow \tan^{-1}(x^2 + 5x + 3) = \tan^{-1}(2x + a)$$

$$\Rightarrow x^2 + 5x + 3 = 2x + a$$

$$\Rightarrow x^2 + 3x + 3 - a = 0$$

$$D \geq 0$$

$$9 - 4(1)(3 - a) \geq 0$$

$$\Rightarrow 9 - 12 + 4a \geq 0$$

$$\Rightarrow 4a \geq 3 \Rightarrow a \geq \frac{3}{4}$$

$$\boxed{a_{\min} = \frac{3}{4}}$$

4. Ans. 1.00

Sol. $f(x) = f(-x)$

$$\frac{2^x - 1}{x^a(2^x + 1)} = \frac{2^{-x} - 1}{(-x)^a(2^{-x} + 1)}$$

$$\Rightarrow \frac{2^x - 1}{x^a(2^x + 1)} = \frac{-(2^x - 1)}{(-x)^a(2^{-x} + 1)}$$

$$\Rightarrow x^a = -(-x)^a$$

5. Ans. 3.00

Sol. $6\{x\} - 3[x] = 2x$

$$\Rightarrow 6\{x\} - 3[x] = 2[x] + 2\{x\}$$

$$\Rightarrow 4\{x\} = 5[x]$$

$$\Rightarrow \frac{5}{4}[x] = \{x\} \quad \dots(1)$$

$$\Rightarrow 0 \leq \{x\} < 1$$

$$\Rightarrow 0 \leq \frac{5}{4}[x] < 1$$

$$\Rightarrow 0 \leq [x] < \frac{4}{5}$$

$$\Rightarrow [x] = 0$$

From equation (1)

$$\{x\} = 0$$

$$\Rightarrow x = 0$$

$$\text{Number of solution} = 1 = N$$

$$\Rightarrow N + 2 = 3$$

6. Ans. 1.00

Sol. L.H.L = $\lim_{x \rightarrow 1^-} g(f(x))$

$$= g(f(1^-))$$

$$= g(1^-) = 2$$

$$\text{R.H.L} = \lim_{x \rightarrow 1^+} g(f(x))$$

$$= g(f(1^+))$$

$$= g(1^+) = a + 1$$

$$\Rightarrow a + 1 = 2 \Rightarrow a = 1$$

7. Ans. 0.25

Sol. $f(x+1) + f(2x+2) + f(1-x) + f(2+x) = x+1$

for $x = 0$

$$f(1) + f(2) + f(1) + f(2) = 1$$

$$\Rightarrow f(1) + f(2) = \frac{1}{2} \quad \dots (1)$$

for $x = -1$

$$f(0) + f(0) + f(2) + f(1) = 0 \quad \dots (2)$$

From eq. (1) and (2)

$$2f(0) + \frac{1}{2} = 0 \Rightarrow f(0) = -\frac{1}{4}$$

$$\Rightarrow |f(0)| = \frac{1}{4} = 0.25 \text{ Ans.}$$

8. Ans. 7.00

Sol. $\sin^{-1}(\sin 8) + \cos^{-1}(\cos 6) + \tan^{-1}(\tan 9)$

$$= \sin^{-1}(\sin(3\pi - 8)) + \cos^{-1}(\cos(2\pi - 6)) + \tan^{-1}(\tan(9 - 3\pi))$$

$$= 3\pi - 8 + 2\pi - 6 + 9 - 3\pi$$

$$= 2\pi - 5$$

$$\Rightarrow a = 2, b = -5$$

$$\text{So, } a - b = 7$$

9. Ans. 5.00

Sol. $f(x) = \sin^{-1}x + |\sin^{-1}x| + \sin^{-1}|x|$

$$\Rightarrow f(x) = \begin{cases} 3\sin^{-1}x & 0 \leq x \leq 1 \\ -\sin^{-1}x & -1 \leq x < 0 \end{cases}$$

$$\text{Range} = \left[0, \frac{3\pi}{2}\right]$$

$$\text{Integers in the interval } \left[0, \frac{3\pi}{2}\right] \text{ are } 0, 1, 2, 3, 4$$

So, No. of integers in the range equal to '5'

10. Ans. 5.00

Sol. $f(-x) = f(x) \quad \dots (1)$

$$\text{and } f(x) = f(x+6)$$

$$\text{Now } f(8) + f(-3)$$

$$= f(2) + f(3)$$

$$= (5-2) + (5-3) = 5$$